SAP ABAP Developer

Capstone Project Report

Design and Implementation of Custom ALV Report Using SAP ABAP

Submitted By	Shivam
Roll Number	23051462
Program	B.TECH in Computer Science Engineering
Institution	KIIT University
Course	SAP ABAP Developer
Topic	Real ABAP Development Scenario - Custom ALV Report
Submission	April 2026

1. Problem Statement

In modern organisations, large volumes of business data are stored in SAP database tables. However, standard SAP reports do not always provide flexible data visualization according to business requirements. Users often need customized reports to view specific fields, apply filters, and analyse data efficiently.

Manual extraction of data from SAP tables is time-consuming and requires technical knowledge. Therefore, organizations require custom-developed reports that present structured information in an easy-to-understand format.

This project focuses on the design and implementation of a **Custom ALV (ABAP List Viewer) Report** for a fictitious company **ABC Technologies Pvt Ltd** using **SAP ABAP programming**.

The report retrieves transactional data from SAP tables and displays it in an interactive ALV Grid format that supports sorting, filtering, and structured output. This solution improves reporting efficiency and demonstrates practical implementation of ABAP development concepts.

2. Company Overview & SAP Data Structure

ABC Technologies Pvt Ltd is a technology-based company that manages large volumes of business data using SAP ERP system.

The organization stores transactional and master data in relational tables. To analyze business information efficiently, the company requires customized SAP reports developed using ABAP programming language.

The following SAP tables are used in this project:

Module	SAP Table	Description
SD	VBAK	Sales Document Header Data
SD	VBAP	Sales Document Item Data
FI	BKPF	Accounting Document Header
M M	MARA	Material Master Data
ABAP	ALV Grid	Structured Report Output

Table 1 – SAP Tables Used in Report Development

3. ALV Report Development (Steps)

The ALV cycle is an eight-step, cross-departmental process. Each step is executed via a dedicated SAP transaction, and data flows automatically between steps – eliminating redundant entry and ensuring consistency across ABAP, ABAP, and ABAP.

Step	Activity	T-Code	Dept	Key Output / SAP Document
1	Create ABAP Program	SE38	ABAP	Z program created
2	Create Package	SE80	ABAP	Development object stored
3	Define Table Structure	SE11	ABAP	SAP table selected
4	Write SELECT Query	SE38	ABAP	Data retrieved from database
5	Create Internal Table	SE38	ABAP	Structured dataset stored
6	Create Field Catalog	SE38	ABAP	Column headings defined
7	Generate ALV Output	SE38	ABAP	Interactive ALV report generated
8	Execute Program	F8	User	Final report output displayed

Table 2 - ALV Process Steps with T-Codes, Departments & Outputs

Report Data Retrieval (SE38)

The Custom ALV Report program is created using transaction **SE38 (ABAP Editor)**. The program retrieves data from SAP database tables such as VBAK (Sales Document Header) using SELECT query statements.

The SELECT statement extracts required fields such as:

- Sales Document Number (VBELN)
- Creation Date (ERDAT)
- Created By (ERNAM)

The retrieved data is stored in an internal table which acts as a temporary storage structure for processing report data.

Internal tables improve performance and allow structured handling of large datasets within ABAP programs.

Internal Table Processing

The internal table is defined to store multiple rows of structured data retrieved from SAP database tables.

Each row of the internal table represents one record from the database.

Work areas are used to process single row data and map fields correctly before displaying the output.

Proper definition of data types ensures accuracy and consistency in report structure.

ALV Output Generation (ALV Grid Display)

The ALV (ABAP List Viewer) Grid is used to display the retrieved data in structured tabular format.

Field Catalogue is created to define:

- column headings
- field order
- display format

The function module **REUSE_ALV_GRID_DISPLAY** is used to generate interactive output.

ALV Grid provides features such as:

- column sorting
- data filtering
- layout adjustment
- structured display
- improved readability

The final output is displayed on SAP GUI screen where users can analyze report data efficiently.

4. SAP Module Integration — ABAP and Database Tables

SAP ABAP programming language interacts directly with SAP database tables to retrieve and process business data.

The ALV report reads structured data from tables and converts it into user-friendly format.

Integration Point	Object	Purpose	Outcome
Data Retrieval	VBAK	Sales document data	report dataset generated
Material Data	MARA	Material details	Structured information displayed
Internal Tables	ABAP	Temporary data storage	Faster processing
ALV Grid	ALV Function	Report visualization	Interactive output

Table 3 – Data Flow in ALV Report

5. Tech Stack & SAP Tools Used

Component	Technology / Tool	Purpose
Programming Language	ABAP	Report development
ERP Platform	SAP S/4HANA	Data storage and processing
Development Tool	SE38	Program creation
Object Navigator	SE80	Package management
Data Dictionary	SE11	Table structure
Database	SAP HANA	Data storage
Report Tool	ALV Grid	Report display
Configuration	SAP IMG	System setup

Table 4 – Technology Stack

6. Business Benefits Impact

Implementation of Custom ALV Report improves efficiency of data analysis and reporting.

KPI	Before (Manual)	After (SAP)	Improvement
Data Retrieval Time	20-30 min	Few seconds	faster access
Report Accuracy	medium	high	Reduced errors
Data Visibility	limited	structured	Better understanding
Manual Effort	high	low	automation
Decision Making	slow	fast	Real-time data

Table 5 – KPI Improvement Matrix

7. Unique Points & Design Decisions

• **Custom ALV Report Development:** The project focuses on development of a custom ABAP report instead of using standard SAP reports, providing flexibility in selecting fields and structuring output format.

• **Structured ALV Grid Layout:** ALV Grid format is used to present report output in structured tabular form with column headings, improving readability and usability of business data.

• **Reusable Program Design:** The ABAP program is designed in modular format so that additional fields or tables can be added easily in future without major code modification.

• **Optimized Data Selection:** The SELECT query retrieves only required fields from SAP database tables, improving performance and reducing system load.

• **Field Catalog Configuration:** Field catalog structure allows customization of column names, order, and display format, enabling flexible report design.

• **Interactive Report Features:** ALV Grid provides built-in functionalities such as sorting, filtering, layout adjustment, and structured display, improving user experience.

• **Technical Implementation Approach:** The project follows step-by-step ABAP development lifecycle including program creation, data retrieval, internal table handling, and ALV output generation.

• **Scalable Report Architecture:** The report design allows integration with additional SAP tables such as VBAP or MARA, enabling advanced reporting capabilities.

8. Challenges & Risk Mitigation

Challenge	Impact	Mitigation & SAP Control
Incorrect table selection	Wrong data	Verified table structure in SE11
Large data volume	Slow performance	Limited rows using SELECT condition
Incorrect field mapping	Display errors	Defined field catalog properly
Syntax errors	Program failure	Debugging in SE38
Output formatting issue	Unclear report	ALV grid structure used

Table 6 – Risk Register with SAP Controls

8b. Future Improvements

• **SAP Fiori Integration:** The ALV report can be converted into a Fiori application using OData services to provide modern web-based user interface accessible from browser and mobile devices.

• **CDS View Integration:** Core Data Services (CDS) views can be used to improve data modelling and performance by pushing data processing to the HANA database layer.

• **Export to Excel Functionality:** Additional functionality can be added to allow users to export ALV report output directly into Excel format for offline analysis and reporting.

• **Selection Screen Enhancement:** Input parameters such as date range, document number, or user name can be added to allow dynamic filtering of report results.

• **Advanced ALV (OO ALV) Implementation:** Object-Oriented ALV (CL_GUI_ALV_GRID) can be used to provide enhanced interactive features and improved performance.

• **Integration with Multiple Tables:** Additional SAP tables such as VBAP (sales item data) or MARA (material master data) can be integrated to provide more detailed reporting.

• **Authorization Control:** Role-based access control can be implemented to restrict report access based on user authorization levels.

• **Graphical Dashboard Integration:** Report data can be visualized using charts and analytics tools for better business insights.

9. SAP Transaction Screenshots

The following screenshots illustrate the key SAP transactions executed during the ALV process for ABC Technologies Pvt Ltd. Each screen reflects the actual SAP GUI interface and the data entered at each step of the cycle.

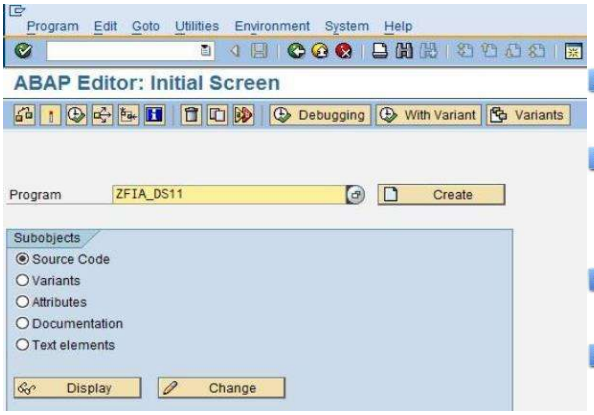


Figure 1 - SE38 Program Creation Screen

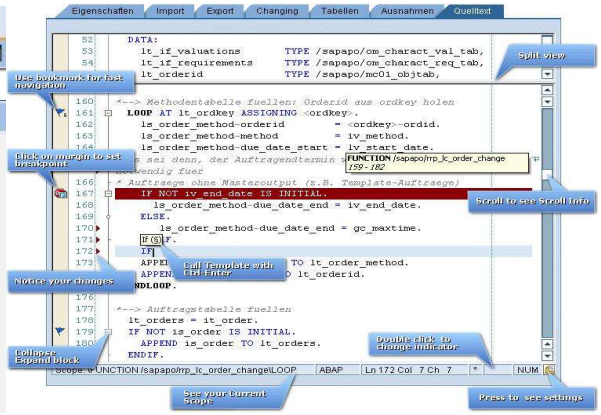
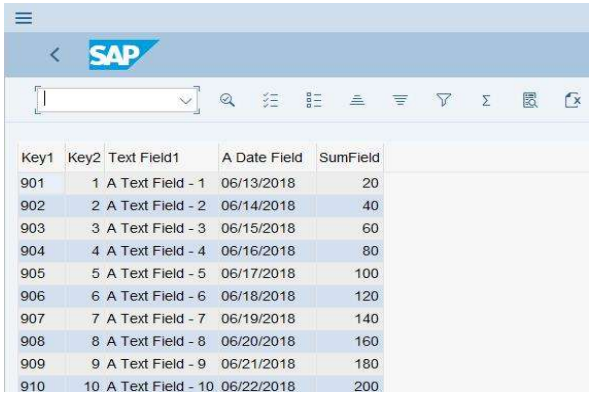


Figure 2 – ABAP Code Editor



Key1	Key2	Text Field1	A Date Field	SumField
901	1	A Text Field - 1	06/13/2018	20
902	2	A Text Field - 2	06/14/2018	40
903	3	A Text Field - 3	06/15/2018	60
904	4	A Text Field - 4	06/16/2018	80
905	5	A Text Field - 5	06/17/2018	100
906	6	A Text Field - 6	06/18/2018	120
907	7	A Text Field - 7	06/19/2018	140
908	8	A Text Field - 8	06/20/2018	160
909	9	A Text Field - 9	06/21/2018	180
910	10	A Text Field - 10	06/22/2018	200

Figure 3 – SELECT Query Implementation

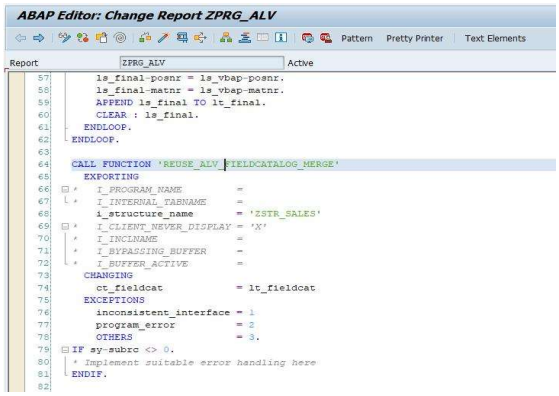
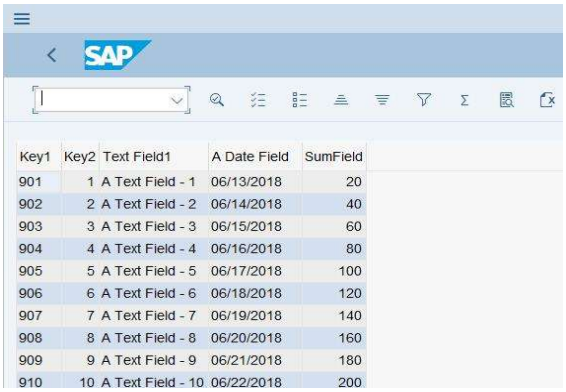


Figure 4 – Field Catalog Structure



Key1	Key2	Text Field1	A Date Field	SumField
901	1	A Text Field - 1	06/13/2018	20
902	2	A Text Field - 2	06/14/2018	40
903	3	A Text Field - 3	06/15/2018	60
904	4	A Text Field - 4	06/16/2018	80
905	5	A Text Field - 5	06/17/2018	100
906	6	A Text Field - 6	06/18/2018	120
907	7	A Text Field - 7	06/19/2018	140
908	8	A Text Field - 8	06/20/2018	160
909	9	A Text Field - 9	06/21/2018	180
910	10	A Text Field - 10	06/22/2018	200

Figure 5 - ALV Grid Output Screen

10. Conclusion

This capstone project demonstrates the design and implementation of a Custom ALV Report using SAP ABAP programming language.

The project covers complete development lifecycle including:

- program creation
- database table selection
- data retrieval using SELECT query
- internal table processing
- ALV grid output generation

The solution improves efficiency of business reporting by providing structured and interactive output format.

The project demonstrates practical knowledge of SAP ABAP development concepts which are essential for real-time SAP technical consultant roles.

The skills gained through this project provide strong foundation for future development in SAP technologies including Fiori, CDS Views, and advanced reporting tools.

Shivam | Roll No: 23051462 | B.Tech CSE – KIIT University | SAP ABAP Developer | April 2026